

## How-To: Lifecycle Hooks

ragent hooks let you run shell commands at key points in the session lifecycle. This guide covers the two mechanisms for controlling tool execution: **PreToolUse** and **PostToolUse** hooks.

### Table of Contents

- Configuration
  - PreToolUse Hooks
    - JSON-Decision Protocol
    - Exit-Code Convention
    - Precedence Rules
  - PostToolUse Hooks
    - Exit-Code Convention
    - Modified Output
  - Other Hook Triggers
  - Environment Variables
  - Complete Example
- 

### Configuration

Hooks are defined in the `hooks` array of `ragent.json`:

```
{
  "hooks": [
    {
      "trigger": "pre_tool_use",
      "command": "/path/to/my-pre-tool-hook.sh",
      "timeout_secs": 30
    },
    {
      "trigger": "post_tool_use",
      "command": "/path/to/my-post-tool-hook.sh",
      "timeout_secs": 10
    }
  ]
}
```

Each hook entry has three fields:

| Field                     | Type    | Default | Description                                                                                                                                                             |
|---------------------------|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>trigger</code>      | string  | —       | When to fire (see triggers)                                                                                                                                             |
| <code>command</code>      | string  | —       | Shell command to execute (runs via <code>sh -c</code> )                                                                                                                 |
| <code>timeout_secs</code> | integer | 30      | Timeout in seconds. Enforced for <code>post_tool_use</code> and the fire-and-forget triggers; <b>not</b> enforced for <code>pre_tool_use</code> (see PreToolUse Hooks). |

Invalid hook entries (for example an unknown `trigger` string) are skipped at load time with a log warning — no startup error surfaces to the user.

---

## PreToolUse Hooks

PreToolUse hooks fire **before** a tool is executed. They can approve, deny, modify, block, or warn about the upcoming tool call.

Two mechanisms control tool execution, and they work side by side:

1. **JSON-Decision Protocol** — the hook writes a JSON decision to stdout.
2. **Exit-Code Convention** — the hook's exit code signals block / warn / error.

### JSON-Decision Protocol

When the hook exits with code **0**, ragent parses stdout as JSON. The following decisions are recognised:

| stdout JSON                                                                                             | Effect                                                                             |
|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <code>{"decision": "allow"}</code>                                                                      | Skip the UI permission prompt and allow the tool.                                  |
| <code>{"decision": "deny", "reason": "..."}<br/>{ "modified_input": { "path":<br/>"new/path" } }</code> | Deny the tool call with an optional reason.<br>Replace the tool's input arguments. |
| <i>(empty or invalid JSON)</i>                                                                          | No decision — normal permission flow applies.                                      |

#### Example — allow a specific tool:

```
#!/bin/sh
# Allow all read operations without prompting
echo '{"decision": "allow"}'
```

#### Example — deny with a reason:

```
#!/bin/sh
# Block writes to /etc
echo '{"decision": "deny", "reason": "Writes to /etc are not allowed"}'
```

#### Example — modify input arguments:

```
#!/bin/sh
# Redirect all file reads to a sandbox directory
echo '{"modified_input": { "path": "/sandbox/src/main.rs" } }'
```

### Exit-Code Convention

When the hook exits with a **non-zero** status code, ragent interprets the exit code **before** parsing stdout JSON:

| Exit code | Behaviour                                       |
|-----------|-------------------------------------------------|
| <b>0</b>  | Parse stdout JSON (see JSON-Decision Protocol). |

| Exit code | Behaviour                                                                                                                                                                |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | <b>Warn</b> — allow the tool to proceed, but emit <code>tracing::warn!</code> and publish <code>Event::HookWarning</code> on the event bus so the TUI can surface it.    |
| 2         | <b>Block</b> — the tool call is blocked. The hook's <code>stderr</code> (trimmed, capped at 500 characters) is used as the block reason. Stdout JSON is <b>ignored</b> . |
| $\geq 3$  | <b>Error</b> — treat as a hook failure (not a policy decision). Fall through to the normal permission flow with a <code>tracing::error!</code> diagnostic.               |

#### Example — block with exit code 2:

```
#!/bin/sh
# Block all bash commands that contain "rm -rf"
if echo "$RAGENT_TOOL_INPUT" | grep -q "rm -rf"; then
    echo "Blocked: destructive command detected" >&2
    exit 2
fi
# Otherwise, allow without prompting
echo '{"decision": "allow"}'
```

#### Example — warn with exit code 1:

```
#!/bin/sh
# Warn when a write tool is used outside src/
if echo "$RAGENT_TOOL_INPUT" | grep -qv '"path":.*"src/'; then
    echo "Warning: write outside src/ directory" >&2
    exit 1
fi
exit 0
```

#### Example — hook error with exit code 3:

```
#!/bin/sh
# This hook has a bug (missing dependency)
if ! command -v jq >/dev/null 2>&1; then
    echo "Hook error: jq is not installed" >&2
    exit 3
fi
# Normal processing would go here
echo '{"decision": "allow"}'
```

When a hook exits with code  $\geq 3$ , the tool call is **not** blocked — it falls through to the normal permission flow so a broken hook cannot lock the user out of all tool use.

#### Precedence Rules

1. **Exit code 2 takes absolute precedence** over stdout JSON. Even if the hook writes `{"decision": "allow"}` to stdout, exit code 2 will block the tool call.
2. **Exit code 1 does not parse stdout JSON**. The warning is emitted regardless of stdout content.
3. **Multiple hooks are evaluated in order**. The first hook that returns a decision (allow, deny, modified\_input, or Blocked) wins; remaining hooks are not executed.

4. **Spawn failure** is treated as exit code  $\geq 3$  (hook error). The tool call falls through to the normal permission flow.

**No timeout on the pre-hook path:** PreToolUse hooks are spawned synchronously via `std::process::Command::output()` with no timeout wrapper, so `timeout_secs` is silently ignored here — a hanging pre-hook blocks every tool call until it exits. `timeout_secs` is enforced for `post_tool_use` hooks and the fire-and-forget triggers.

---

## PostToolUse Hooks

PostToolUse hooks fire **after** a tool has executed. They can modify the tool's output, warn about the result, or flag it as policy-violated.

### Exit-Code Convention

| Exit code                  | Behaviour                                                                                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>0</b>                   | Parse stdout JSON for <code>modified_output</code> (see below).                                                                                                                                                            |
| <b>1</b>                   | <b>Warn</b> — emit <code>tracing::warn!</code> and publish <code>Event::HookWarning</code> . The tool result is not modified.                                                                                              |
| <b>2</b>                   | <b>Flag</b> — publish <code>Event::ToolResultFlagged</code> with <code>stderr</code> as the reason. The tool result is <b>not</b> suppressed (the call already executed), but the flag appears in the session log and TUI. |
| <b><math>\geq 3</math></b> | <b>Error</b> — treat as a hook failure. No effect on the tool result.                                                                                                                                                      |

### Modified Output

When a PostToolUse hook exits with code 0 and writes JSON to stdout, ragent looks for a `modified_output` field:

```
{
  "modified_output": {
    "content": "Sanitised output here",
    "metadata": {"custom": "value"}
  }
}
```

If the `modified_output.content` field is present, it replaces the tool's output content.

#### Example — redact secrets from tool output:

```
#!/bin/sh
# Read the original output from the environment variable
OUTPUT="$RAGENT_TOOL_OUTPUT"
# Redact API keys (simplified example)
REDACTED=$(echo "$OUTPUT" | sed 's/sk-[a-zA-Z0-9]*/[REDACTED]/g')
# Return modified output
echo "{\"modified_output\": {\"content\": \"$REDACTED\"}}"
```

#### Example — flag suspicious output:

```
#!/bin/sh
# Check if the tool output contains sensitive data
if echo "$RAGENT_TOOL_OUTPUT" | grep -q "password"; then
    echo "Flagged: output contains potential password" >&2
    exit 2
fi
exit 0
```

---

## Other Hook Triggers

In addition to `pre_tool_use` and `post_tool_use`, `ragent` supports these triggers:

| Trigger                           | Description                                                  |
|-----------------------------------|--------------------------------------------------------------|
| <code>on_session_start</code>     | Fired when a session receives its first user message.        |
| <code>on_session_end</code>       | Fired after a session completes processing a user message.   |
| <code>on_error</code>             | Fired when an LLM call or tool execution returns an error.   |
| <code>on_permission_denied</code> | Fired when a tool call is rejected due to a permission rule. |
| <code>on_turn_start</code>        | Fired at the start of each agent turn/iteration.             |
| <code>on_turn_end</code>          | Fired at the end of each agent turn/iteration.               |
| <code>on_compaction</code>        | Fired when context compaction is performed.                  |

---

These triggers fire asynchronously — errors are logged but never fatal. The `RAGENT_ERROR` environment variable is set for `on_error` hooks.

---

## Environment Variables

All hooks receive these environment variables:

| Variable                        | Description                                        |
|---------------------------------|----------------------------------------------------|
| <code>RAGENT_TRIGGER</code>     | The trigger name (e.g. <code>pre_tool_use</code> ) |
| <code>RAGENT_WORKING_DIR</code> | The session working directory                      |

---

`on_error` hooks additionally receive:

| Variable                  | Description       |
|---------------------------|-------------------|
| <code>RAGENT_ERROR</code> | The error message |

---

`PreToolUse` hooks additionally receive:

| Variable          | Description                        |
|-------------------|------------------------------------|
| RAGENT_TOOL_NAME  | The name of the tool being invoked |
| RAGENT_TOOL_INPUT | JSON string of the tool arguments  |

PostToolUse hooks additionally receive:

| Variable            | Description                           |
|---------------------|---------------------------------------|
| RAGENT_TOOL_NAME    | The name of the tool that was invoked |
| RAGENT_TOOL_INPUT   | JSON string of the tool arguments     |
| RAGENT_TOOL_OUTPUT  | JSON string of the tool output        |
| RAGENT_TOOL_SUCCESS | "true" or "false"                     |

Turn and compaction triggers additionally receive:

| Variable                       | Description                                                      |
|--------------------------------|------------------------------------------------------------------|
| RAGENT_TURN_NUMBER             | Current agent turn/iteration number (on_turn_start, on_turn_end) |
| RAGENT_COMPACTON_REASON        | Why compaction fired (on_compaction)                             |
| RAGENT_COMPACTON_TOKENS_BEFORE | Token count before compaction (on_compaction)                    |
| RAGENT_COMPACTON_TOKENS_AFTER  | Token count after compaction (on_compaction)                     |

A **plugin-contributed** hook (see `plugins.md` section 9.6) additionally receives:

| Variable           | Description                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------|
| CLAUDE_PLUGIN_ROOT | The declaring plugin's root, so a command can resolve <code>\${CLAUDE_PLUGIN_ROOT}/...</code> |
| CLAUDE_PROJECT_DIR | Alias of <code>RAGENT_WORKING_DIR</code> , for Claude-dialect scripts                         |

and the event as JSON on **stdin** (the Claude hook protocol shape: `hook_event_name`, `cwd`, `session_id`, and — for the tool-scoped triggers — `tool_name`, `tool_input`, `tool_response`, `tool_success`). A hook declared in `ragent.json` keeps the environment-only contract (no stdin payload).

## Stop Hook Continuation

The `on_session_end` trigger doubles as the Claude **Stop** hook. A Stop hook may block the turn and return findings (the Claude `asyncRewake` shape):

- `exit 2` with a stdout JSON `reason/additionalContext`, else the `stderr` text;
- `exit 0` with `hookSpecificOutput.additionalContext`; or
- a top-level `{"decision": "block", ...}`.

When a Stop hook returns non-empty findings they are appended to the conversation as a synthetic user turn (with an AgentNotice) **before the turn ends**, so a plugin security review can feed corrections back to the model. The loop is bounded to three continuations (`MAX_STOP_CONTINUATIONS = 3`), so a hook that always blocks cannot spin the session.

---

## Complete Example

Here is a `ragent.json` snippet that configures both `PreToolUse` and `PostToolUse` hooks:

```
{
  "hooks": [
    {
      "trigger": "pre_tool_use",
      "command": ".ragent/hooks/pre-tool-use.sh",
      "timeout_secs": 10
    },
    {
      "trigger": "post_tool_use",
      "command": ".ragent/hooks/post-tool-use.sh",
      "timeout_secs": 10
    }
  ]
}
```

**.ragent/hooks/pre-tool-use.sh:**

```
#!/bin/sh
# PreToolUse hook: enforce security policies

TOOL="$RAGENT_TOOL_NAME"
INPUT="$RAGENT_TOOL_INPUT"

# Block bash commands that contain "rm -rf /"
if [ "$TOOL" = "bash" ] && echo "$INPUT" | grep -q "rm -rf /"; then
  echo "Blocked: refused to execute destructive command" >&2
  exit 2
fi

# Warn about writes outside the project directory
if [ "$TOOL" = "write" ] && ! echo "$INPUT" | grep -q '"path":.*src/'; then
  echo "Warning: write target is outside src/" >&2
  exit 1
fi

# Allow all other tools without prompting
echo '{"decision": "allow"}'
```

**.ragent/hooks/post-tool-use.sh:**

```
#!/bin/sh
```

```

# PostToolUse hook: audit tool results

OUTPUT="$RAGENT_TOOL_OUTPUT"

# Flag outputs that contain potential secrets
if echo "$OUTPUT" | grep -qE "(sk-[a-zA-Z0-9]{20,}|AKIA[A-Z0-9]{16})"; then
    echo "Flagged: output contains potential API key" >&2
    exit 2
fi

# No modification needed
exit 0

```

---

## Summary

| Mechanism     | PreToolUse                          | PostToolUse                           |
|---------------|-------------------------------------|---------------------------------------|
| Exit 0        | Parse stdout JSON for decision      | Parse stdout JSON for modified_output |
| Exit 1        | Warn + allow (publish HookWarning)  | Warn (publish HookWarning)            |
| Exit 2        | Block (publish denial to agent)     | Flag (publish ToolResultFlagged)      |
| Exit ≥ 3      | Hook error → normal permission flow | Hook error → no effect on result      |
| Spawn failure | Treated as exit ≥ 3                 | Treated as exit ≥ 3                   |
| Timeout       | Not enforced (no timeout wrapper)   | Treated as exit ≥ 3 (timeout_secs)    |

When multiple PostToolUse hooks fire, the aggregated outcome follows the precedence **Flagged** > **Warn** > **Ok**, and among Ok results the last modified\_output wins.

The two mechanisms (JSON decisions and exit codes) are **complementary**: JSON decisions provide fine-grained control (allow / deny / modify), while exit codes provide a simple, language-agnostic way to signal block, warn, or error from any scripting language.